

Integrated AI Platform

Installation & Deployment Guide

دليل التثبيت والتشغيل التقني للمنصة

Graduation Project | Faculty of Computer Science

THE FOUR CORE MODULES



Digital Auction

Qassim's premier digital bidding portal for local commerce.



Malaria AI

Neural network detection system for medical microscopy images.



Admin Dashboard

Full-scale monitoring of users, bids, and system health.



Sales Analytics

Engineering system for sales data and predictive modeling.

TECHNICAL PREREQUISITES

3.11

Python Version

Environment Compliance

For AI libraries compatibility (TensorFlow & Keras):

- ✓ Python 3.10 or 3.11 is **Required**.
- ✗ Python 3.13 is currently incompatible with Keras layers.
- 🛡️ Use Virtual Environments (venv) for isolation.

Phase 1:

Environment Setup

Initialize your workspace by creating an isolated environment to prevent library conflicts.

```
python -m venv .venv  
.venv\Scripts\activate
```

هذه الخطوة تضمن استقرار المكتبات المستخدمة في المشروع.



PHASE 2: INSTALLING DEPENDENCIES

Execute the batch installation command from the root directory containing `manage.py`.

```
pip install -r requirements.txt
```

Libraries included:

Django 4.2+

TensorFlow

Pandas

Pillow

-  Automated library detection.
-  Version-specific AI modules.
-  Requirements tracking.

PHASE 3: MIGRATION & AUTH

1. Migrations

Prepare the SQL schema for auctions, profiles, and analytics.

```
python manage.py makemigrations
```

2. Database Apply

Apply tables to the local SQLite/PostgreSQL database.

```
python manage.py migrate
```

3. Administrator

Create a superuser for dashboard and portal access.

```
python manage.py  
createsuperuser
```

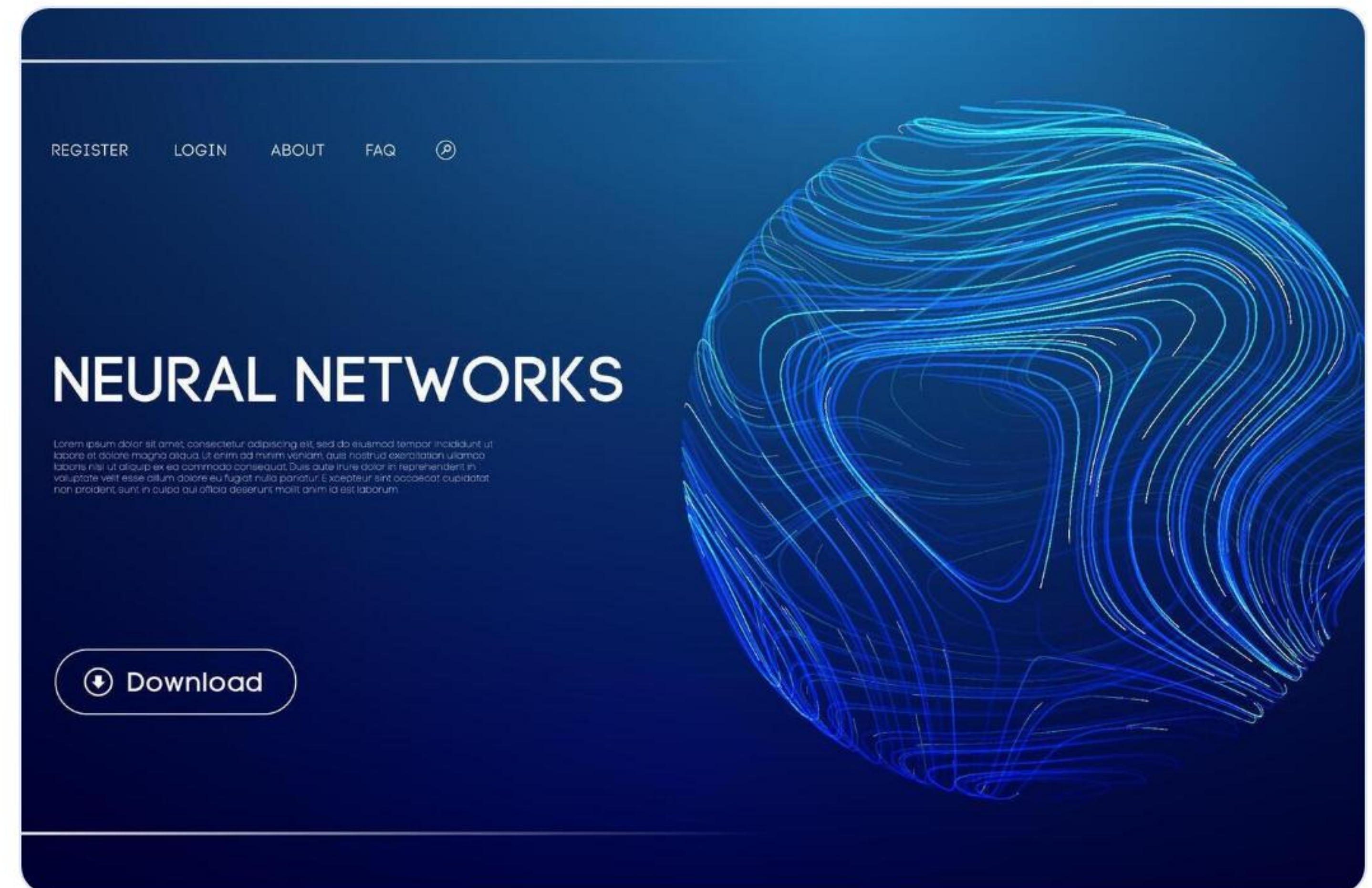

PHASE 4: LAUNCH SEQUENCE

Start the Local Development Server:

```
python manage.py runserver
```

🔗 Access the Portal at:

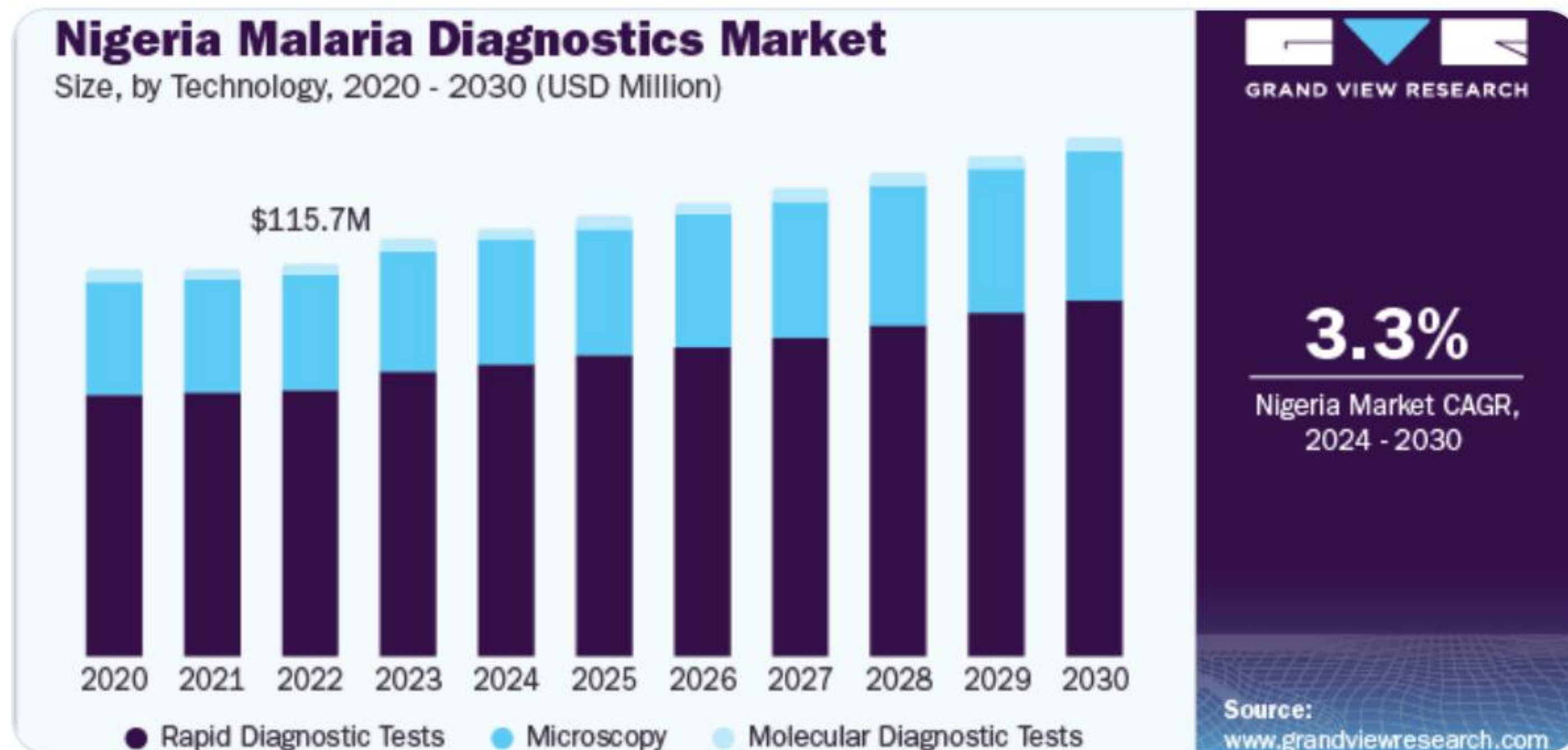
<http://127.0.0.1:8000/>



PLATFORM NAVIGATION MAP

Module / System	URL Path Extension	Functionality
Integrated Portal	/	Gateway to all subsystems
Qassim Auction	/auction-landing/	E-commerce & Bidding UI
Malaria Detection	/malaria/	Image Upload & AI Inference
System Dashboard	/qassim-dashboard/	Metrics & User Management
Sales Engineering	/sales-analysis/	Data Analysis Toolset

VISUAL SHOWCASE



Subsystem Visualization

AI Inference & Data Analysis Modules

- Keep `DEBUG = True` for local static/media serving.
- Ensure the `/media/` folder exists for auction images.
- AI models are loaded on first request - please allow 3-5s for warm-up.

TROUBLESHOOTING GUIDE

⚠ Common Errors

Module Not Found: Re-run `pip install` inside the activated venv.

TensorFlow Error: Check Python version. 3.12+ might cause installation loops.

🖼 Media Issues

If images are missing, check `MEDIA_ROOT` in `settings.py` and ensure the folder is not in `.gitignore`.

Confirm `config/urls.py` includes media pattern helpers.